

Name: Amsalu Tessema Adgeh

ID Number: 2026A8000929002

Course: Data Science

Assignment: DS-A1 | Data Provenance and Measurement Audit

Result Summary Report

This project presents a data provenance and measurement audit of the UCI Bank Marketing dataset (`bankfull.csv`). The research question is: how does the observed term-deposit subscription rate differ between clients contacted by cellular telephone and clients contacted by telephone? The analysis treats each campaign record as the unit of analysis and uses `y` as the subscription outcome. The analysis is descriptive and associational and does not estimate a causal effect.

The frozen dataset contains 45,211 observations and 17 variables. Its SHA-256 hash is `d1513ec63b385506f7cfce9f2c5caa9fe99e7ba4e8c3fa264b3aaf0f849ed32d`. Schema, range, missingness, duplicate, anomaly, and leakage audits were executed. No exact duplicate rows were found, and the defined range checks identified no violations. The audit also identified `unknown` values in several categorical variables, indicating that some unavailable information is represented explicitly rather than as conventional missing values.

The observed subscription rate was 14.92% for cellular contacts and 13.42% for telephone contacts, producing an observed difference of 1.50 percentage points. This difference represents an association within the analyzed records and should not be interpreted as evidence that cellular communication causes higher subscription rates.

The leakage audit identified `duration` as a potential post-contact variable. It would not be available at the beginning of a pre-contact prediction task and therefore could introduce temporal information leakage if used in such a model. The analysis also recognizes that communication channel was not established as randomly assigned, so other characteristics may contribute to the observed difference.

The notebook records the environment, seed, dataset hash, audit results, provenance lineage, and reproducibility information. The final repository preserves the notebook, PDF report, raw dataset, generated results, documentation, and AI-use and independent-verification log.